

Dataset Link -

- <https://www.kaggle.com/datasets/samuelcortinhas/sports-balls-multiclass-image-classification/data>

Connecting Kaggle json

```
In [1]: !mkdir -p ~/.kaggle
        !cp kaggle.json ~/.kaggle/
```

Getting the DataSet

```
In [2]: !kaggle datasets download -d samuelcortinhas/sports-balls-multiclass-image-classifi

Warning: Your Kaggle API key is readable by other users on this system! To fix this,
you can run 'chmod 600 /root/.kaggle/kaggle.json'
Dataset URL: https://www.kaggle.com/datasets/samuelcortinhas/sports-balls-multiclass
-image-classification
License(s): CC0-1.0
Downloading sports-balls-multiclass-image-classification.zip to /content
100% 400M/401M [00:00<00:00, 440MB/s]
100% 401M/401M [00:00<00:00, 540MB/s]
```

Unzipping the data

```
In [3]: !unzip /content/sports-balls-multiclass-image-classification.zip -d /content/
```


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```
inflating: /content/train/volleyball/volleyball_883.jpg
inflating: /content/train/volleyball/volleyball_884.jpg
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inflating: /content/train/volleyball/volleyball_886.jpg
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inflating: /content/train/volleyball/volleyball_888.jpg
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inflating: /content/train/volleyball/volleyball_96.jpg
inflating: /content/train/volleyball/volleyball_97.jpg
inflating: /content/train/volleyball/volleyball_98.jpg
inflating: /content/train/volleyball/volleyball_99.jpg
```

Importing Packages

```
In [4]: import tensorflow as tf
        from tensorflow.keras import layers, models
        from tensorflow.keras.layers import Rescaling
```

```
import tensorflow as tf
import os

# For reproducibility
import numpy as np
import random
import os
import glob
import matplotlib.pyplot as plt
import seaborn as sns

seed = 123
random.seed(seed)
np.random.seed(seed)
tf.random.set_seed(seed)
```

```
In [5]: import os
        import seaborn as sns
        import matplotlib.pyplot as plt
        import pandas as pd

        # Directory path (update with the correct path to your 'train' directory)
        train_dir = '/content/train'

        # Dictionary to store subdirectory names and file counts
        data = {'Category': [], 'Image Count': []}

        # Loop through each subdirectory
        for sub_dir in os.listdir(train_dir):
```

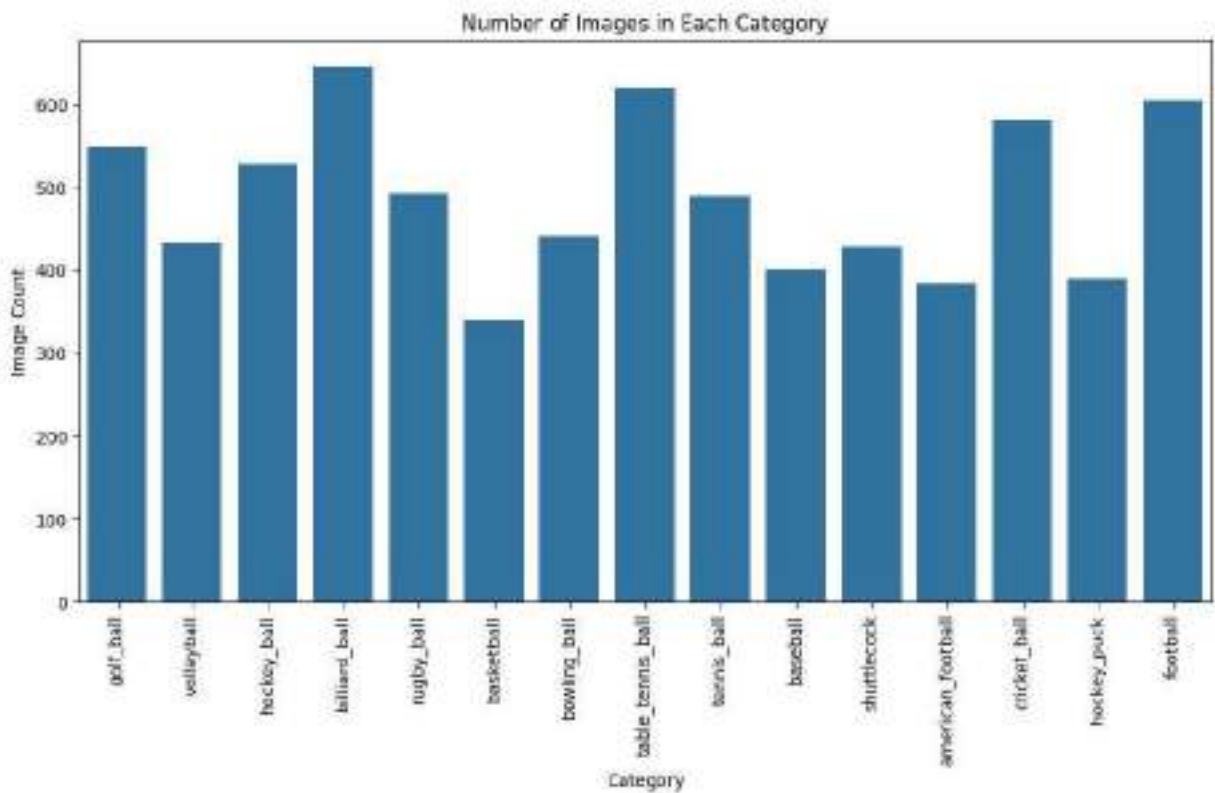
```

sub_dir_path = os.path.join(train_dir, sub_dir)
if os.path.isdir(sub_dir_path):
    # Count the number of files (images) in the subdirectory
    num_files = len([f for f in os.listdir(sub_dir_path) if os.path.isfile(os.p
data['Category'].append(sub_dir)
data['Image Count'].append(num_files)

# Convert the data to a pandas DataFrame
df = pd.DataFrame(data)

# Create the bar plot using Seaborn
plt.figure(figsize=(12, 6))
sns.barplot(x='Category', y='Image Count', data=df)
plt.xticks(rotation=90)
plt.title('Number of Images in Each Category')
plt.ylabel('Image Count')
plt.xlabel('Category')
plt.show()

```



```

In [6]: import os
import random
import matplotlib.pyplot as plt
import matplotlib.image as mpimg

# Directory path (update with the correct path to your 'train' directory)
train_dir = '/content/train'

# Get all subdirectories and files in the directories
all_images = []
for sub_dir in os.listdir(train_dir):
    sub_dir_path = os.path.join(train_dir, sub_dir)
    if os.path.isdir(sub_dir_path):

```



```

        all_images.extend([os.path.join(sub_dir_path, f) for f in os.listdir(sub_dir_path)])

# Randomly select 9 images
random_images = random.sample(all_images, 9)

# Plotting the images in a 3x3 grid
plt.figure(figsize=(10, 10))
for i, image_path in enumerate(random_images):
    img = mpimg.imread(image_path)
    plt.subplot(3, 3, i+1)
    plt.imshow(img)
    plt.axis('off') # Turn off the axis

plt.suptitle('Randomly Selected 9 Images from Directories', fontsize=16)
plt.tight_layout()
plt.show()

```

Randomly Selected 9 Images from Directories



Data Preparation

```
In [7]: # Set up directory and parameters
train_dir = '/content/train'
img_size = (224, 224) # ResNet50 standard input size
batch_size = 32 # Adjust batch size as per your GPU capacity
```

```
In [8]: # Data Augmentation Layers (applied during training)
data_augmentation = tf.keras.Sequential([
    layers.RandomFlip('horizontal'), # Random horizontal flip
    layers.RandomRotation(0.1), # Random rotation
    layers.RandomZoom(0.1), # Random zoom
])
```

Flow from direcotory.

```
In [9]: # Load training and validation datasets using image_dataset_from_directory
train_dataset = tf.keras.preprocessing.image_dataset_from_directory(
    train_dir,
    labels = 'inferred',
    label_mode = 'categorical',
    validation_split=0.2, # 80/20 split for training/validation
    subset="training",
    seed=123, # For reproducibility
    image_size=img_size,
    batch_size=batch_size
)
```

Found 7328 files belonging to 15 classes.
Using 5863 files for training.

```
In [10]: validation_dataset = tf.keras.preprocessing.image_dataset_from_directory(
    train_dir,
    validation_split=0.2,
    labels = 'inferred',
    label_mode = 'categorical',
    subset="validation",
    seed=123,
    image_size=img_size,
    batch_size=batch_size
)
```

Found 7328 files belonging to 15 classes.
Using 1465 files for validation.

```
In [11]: # Apply data augmentation only to the training data
train_dataset = train_dataset.map(lambda x, y: (data_augmentation(x, training=True)
```

```
In [12]: # Optional: Prefetch data for performance improvement
AUTOTUNE = tf.data.experimental.AUTOTUNE
train_dataset = train_dataset.prefetch(buffer_size=AUTOTUNE)
validation_dataset = validation_dataset.prefetch(buffer_size=AUTOTUNE)
```

Model Buidling

```
In [13]: # model = sequeantial ()

# model.add(conv2d)
# model.add(activation)
# model.add(conv2d)
# model.add(activaiton)...

# model.summary()

# x =. sequential()

# x = cov2d()(x)
# x = activation()(x)
```

```
In [14]: def residual_block(x, filters, kernel_size=3, stride=1):
    shortcut = x

    # First 1x1 Convolution to reduce dimensions
    x = layers.Conv2D(filters, kernel_size=1, strides=stride, padding='same')(x)
    x = layers.BatchNormalization()(x)
    x = layers.Activation('relu')(x)

    # Second 3x3 Convolution
    x = layers.Conv2D(filters, kernel_size=kernel_size, strides=1, padding='same')(x)
    x = layers.BatchNormalization()(x)
    x = layers.Activation('relu')(x)

    # Third 1x1 Convolution to restore dimensions
    x = layers.Conv2D(filters * 4, kernel_size=1, strides=1, padding='same')(x)
    x = layers.BatchNormalization()(x)

    # Shortcut connection (adjust dimensions if necessary)
    if stride != 1 or shortcut.shape[-1] != filters * 4:
        shortcut = layers.Conv2D(filters * 4, kernel_size=1, strides=stride, padding='same')(x)
        shortcut = layers.BatchNormalization()(shortcut)

    # Add the shortcut
    x = layers.Add()(x, shortcut)
    x = layers.Activation('relu')(x)
    return x
```

```
In [15]: def build_resnet50(input_shape=(224, 224, 3), num_classes=15):
    inputs = tf.keras.Input(shape=input_shape)

    # x = layers.rescaling(1/255.)
    # Initial Convolution and MaxPooling
    x = layers.Conv2D(64, kernel_size=7, strides=2, padding='same')(inputs)
    x = layers.BatchNormalization()(x)
```

```

x = layers.Activation('relu')(x)
x = layers.MaxPooling2D(pool_size=3, strides=2, padding='same')(x)

# Residual Blocks (Bottleneck Architecture)
x = residual_block(x, filters=64)
x = residual_block(x, filters=64)
x = residual_block(x, filters=64)

x = residual_block(x, filters=128, stride=2)
x = residual_block(x, filters=128)
x = residual_block(x, filters=128)
x = residual_block(x, filters=128)

x = residual_block(x, filters=256, stride=2)
x = residual_block(x, filters=256)
x = residual_block(x, filters=256)
x = residual_block(x, filters=256)
x = residual_block(x, filters=256)
x = residual_block(x, filters=256)

x = residual_block(x, filters=512, stride=2)
x = residual_block(x, filters=512)
x = residual_block(x, filters=512)

# Global Average Pooling and Output
x = layers.GlobalAveragePooling2D()(x)
x = layers.Dense(num_classes, activation='softmax')(x)

model = models.Model(inputs, x)
return model

```

```

In [15]: # Build the model
model = build_resnet50(input_shape=(224, 224, 3))
model.summary()

```

Model: "functional_1"

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0	-
conv2d (Conv2D)	(None, 112, 112, 64)	9,472	input_layer_1[0]...
batch_normalization (BatchNormalizatio...	(None, 112, 112, 64)	256	conv2d[0][0]
activation (Activation)	(None, 112, 112, 64)	0	batch_normalizat...
max_pooling2d (MaxPooling2D)	(None, 56, 56, 64)	0	activation[0][0]
conv2d_1 (Conv2D)	(None, 56, 56, 64)	4,160	max_pooling2d[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2d_1[0][0]
activation_1 (Activation)	(None, 56, 56, 64)	0	batch_normalizat...
conv2d_2 (Conv2D)	(None, 56, 56, 64)	36,928	activation_1[0][...
batch_normalizatio... (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2d_2[0][0]
activation_2 (Activation)	(None, 56, 56, 64)	0	batch_normalizat...
conv2d_3 (Conv2D)	(None, 56, 56, 256)	16,640	activation_2[0][...
conv2d_4 (Conv2D)	(None, 56, 56, 256)	16,640	max_pooling2d[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 56, 56, 256)	1,024	conv2d_3[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 56, 56, 256)	1,024	conv2d_4[0][0]
add (Add)	(None, 56, 56, 256)	0	batch_normalizat... batch_normalizat...
activation_3 (Activation)	(None, 56, 56, 256)	0	add[0][0]
conv2d_5 (Conv2D)	(None, 56, 56, 64)	16,448	activation_3[0][...

batch_normalizatio... (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2d_5[0][0]
activation_4 (Activation)	(None, 56, 56, 64)	0	batch_normalizat...
conv2d_6 (Conv2D)	(None, 56, 56, 64)	36,928	activation_4[0][...
batch_normalizatio... (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2d_6[0][0]
activation_5 (Activation)	(None, 56, 56, 64)	0	batch_normalizat...
conv2d_7 (Conv2D)	(None, 56, 56, 256)	16,640	activation_5[0][...
batch_normalizatio... (BatchNormalizatio...	(None, 56, 56, 256)	1,024	conv2d_7[0][0]
add_1 (Add)	(None, 56, 56, 256)	0	batch_normalizat... activation_3[0][...
activation_6 (Activation)	(None, 56, 56, 256)	0	add_1[0][0]
conv2d_8 (Conv2D)	(None, 56, 56, 64)	16,448	activation_6[0][...
batch_normalizatio... (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2d_8[0][0]
activation_7 (Activation)	(None, 56, 56, 64)	0	batch_normalizat...
conv2d_9 (Conv2D)	(None, 56, 56, 64)	36,928	activation_7[0][...
batch_normalizatio... (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2d_9[0][0]
activation_8 (Activation)	(None, 56, 56, 64)	0	batch_normalizat...
conv2d_10 (Conv2D)	(None, 56, 56, 256)	16,640	activation_8[0][...
batch_normalizatio... (BatchNormalizatio...	(None, 56, 56, 256)	1,024	conv2d_10[0][0]
add_2 (Add)	(None, 56, 56, 256)	0	batch_normalizat... activation_6[0][...
activation_9	(None, 56, 56,	0	add_2[0][0]

(Activation)	256)		
conv2d_11 (Conv2D)	(None, 28, 28, 128)	32,896	activation_9[0][...
batch_normalizatio... (BatchNormalizatio...	(None, 28, 28, 128)	512	conv2d_11[0][0]
activation_10 (Activation)	(None, 28, 28, 128)	0	batch_normalizat...
conv2d_12 (Conv2D)	(None, 28, 28, 128)	147,584	activation_10[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 28, 28, 128)	512	conv2d_12[0][0]
activation_11 (Activation)	(None, 28, 28, 128)	0	batch_normalizat...
conv2d_13 (Conv2D)	(None, 28, 28, 512)	66,048	activation_11[0]...
conv2d_14 (Conv2D)	(None, 28, 28, 512)	131,584	activation_9[0][...
batch_normalizatio... (BatchNormalizatio...	(None, 28, 28, 512)	2,048	conv2d_13[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 28, 28, 512)	2,048	conv2d_14[0][0]
add_3 (Add)	(None, 28, 28, 512)	0	batch_normalizat... batch_normalizat...
activation_12 (Activation)	(None, 28, 28, 512)	0	add_3[0][0]
conv2d_15 (Conv2D)	(None, 28, 28, 128)	65,664	activation_12[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 28, 28, 128)	512	conv2d_15[0][0]
activation_13 (Activation)	(None, 28, 28, 128)	0	batch_normalizat...
conv2d_16 (Conv2D)	(None, 28, 28, 128)	147,584	activation_13[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 28, 28, 128)	512	conv2d_16[0][0]
activation_14 (Activation)	(None, 28, 28, 128)	0	batch_normalizat...

conv2d_17 (Conv2D)	(None, 28, 28, 512)	66,048	activation_14[0]...
batch_normalization_17 (Batch Normalization)	(None, 28, 28, 512)	2,048	conv2d_17[0][0]
add_4 (Add)	(None, 28, 28, 512)	0	batch_normalization_17[0][0]
activation_15 (Activation)	(None, 28, 28, 512)	0	add_4[0][0]
conv2d_18 (Conv2D)	(None, 28, 28, 128)	65,664	activation_15[0]...
batch_normalization_18 (Batch Normalization)	(None, 28, 28, 128)	512	conv2d_18[0][0]
activation_16 (Activation)	(None, 28, 28, 128)	0	batch_normalization_18[0][0]
conv2d_19 (Conv2D)	(None, 28, 28, 128)	147,584	activation_16[0]...
batch_normalization_19 (Batch Normalization)	(None, 28, 28, 128)	512	conv2d_19[0][0]
activation_17 (Activation)	(None, 28, 28, 128)	0	batch_normalization_19[0][0]
conv2d_20 (Conv2D)	(None, 28, 28, 512)	66,048	activation_17[0]...
batch_normalization_20 (Batch Normalization)	(None, 28, 28, 512)	2,048	conv2d_20[0][0]
add_5 (Add)	(None, 28, 28, 512)	0	batch_normalization_20[0][0]
activation_18 (Activation)	(None, 28, 28, 512)	0	add_5[0][0]
conv2d_21 (Conv2D)	(None, 28, 28, 128)	65,664	activation_18[0]...
batch_normalization_21 (Batch Normalization)	(None, 28, 28, 128)	512	conv2d_21[0][0]
activation_19 (Activation)	(None, 28, 28, 128)	0	batch_normalization_21[0][0]
conv2d_22 (Conv2D)	(None, 28, 28, 128)	147,584	activation_19[0]...
batch_normalization_22 (Batch Normalization)	(None, 28, 28, 128)	512	conv2d_22[0][0]

activation_20 (Activation)	(None, 28, 28, 128)	0	batch_normalizat...
conv2d_23 (Conv2D)	(None, 28, 28, 512)	66,048	activation_20[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 28, 28, 512)	2,048	conv2d_23[0][0]
add_6 (Add)	(None, 28, 28, 512)	0	batch_normalizat... activation_18[0]...
activation_21 (Activation)	(None, 28, 28, 512)	0	add_6[0][0]
conv2d_24 (Conv2D)	(None, 14, 14, 256)	131,328	activation_21[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_24[0][0]
activation_22 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_25 (Conv2D)	(None, 14, 14, 256)	590,080	activation_22[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_25[0][0]
activation_23 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_26 (Conv2D)	(None, 14, 14, 1024)	263,168	activation_23[0]...
conv2d_27 (Conv2D)	(None, 14, 14, 1024)	525,312	activation_21[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv2d_26[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv2d_27[0][0]
add_7 (Add)	(None, 14, 14, 1024)	0	batch_normalizat... batch_normalizat...
activation_24 (Activation)	(None, 14, 14, 1024)	0	add_7[0][0]
conv2d_28 (Conv2D)	(None, 14, 14, 256)	262,400	activation_24[0]...
batch_normalizatio...	(None, 14, 14,	1,024	conv2d_28[0][0]

(BatchNormalizatio...	256)		
activation_25 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_29 (Conv2D)	(None, 14, 14, 256)	590,080	activation_25[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_29[0][0]
activation_26 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_30 (Conv2D)	(None, 14, 14, 1024)	263,168	activation_26[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv2d_30[0][0]
add_8 (Add)	(None, 14, 14, 1024)	0	batch_normalizat... activation_24[0]...
activation_27 (Activation)	(None, 14, 14, 1024)	0	add_8[0][0]
conv2d_31 (Conv2D)	(None, 14, 14, 256)	262,400	activation_27[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_31[0][0]
activation_28 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_32 (Conv2D)	(None, 14, 14, 256)	590,080	activation_28[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_32[0][0]
activation_29 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_33 (Conv2D)	(None, 14, 14, 1024)	263,168	activation_29[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv2d_33[0][0]
add_9 (Add)	(None, 14, 14, 1024)	0	batch_normalizat... activation_27[0]...
activation_30 (Activation)	(None, 14, 14, 1024)	0	add_9[0][0]

conv2d_34 (Conv2D)	(None, 14, 14, 256)	262,400	activation_30[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_34[0][0]
activation_31 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_35 (Conv2D)	(None, 14, 14, 256)	590,080	activation_31[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_35[0][0]
activation_32 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_36 (Conv2D)	(None, 14, 14, 1024)	263,168	activation_32[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv2d_36[0][0]
add_10 (Add)	(None, 14, 14, 1024)	0	batch_normalizat... activation_30[0]...
activation_33 (Activation)	(None, 14, 14, 1024)	0	add_10[0][0]
conv2d_37 (Conv2D)	(None, 14, 14, 256)	262,400	activation_33[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_37[0][0]
activation_34 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_38 (Conv2D)	(None, 14, 14, 256)	590,080	activation_34[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_38[0][0]
activation_35 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_39 (Conv2D)	(None, 14, 14, 1024)	263,168	activation_35[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv2d_39[0][0]
add_11 (Add)	(None, 14, 14, 1024)	0	batch_normalizat... activation_33[0]...

activation_36 (Activation)	(None, 14, 14, 1024)	0	add_11[0][0]
conv2d_40 (Conv2D)	(None, 14, 14, 256)	262,400	activation_36[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_40[0][0]
activation_37 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_41 (Conv2D)	(None, 14, 14, 256)	590,080	activation_37[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv2d_41[0][0]
activation_38 (Activation)	(None, 14, 14, 256)	0	batch_normalizat...
conv2d_42 (Conv2D)	(None, 14, 14, 1024)	263,168	activation_38[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv2d_42[0][0]
add_12 (Add)	(None, 14, 14, 1024)	0	batch_normalizat... activation_36[0]...
activation_39 (Activation)	(None, 14, 14, 1024)	0	add_12[0][0]
conv2d_43 (Conv2D)	(None, 7, 7, 512)	524,800	activation_39[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv2d_43[0][0]
activation_40 (Activation)	(None, 7, 7, 512)	0	batch_normalizat...
conv2d_44 (Conv2D)	(None, 7, 7, 512)	2,359,808	activation_40[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv2d_44[0][0]
activation_41 (Activation)	(None, 7, 7, 512)	0	batch_normalizat...
conv2d_45 (Conv2D)	(None, 7, 7, 2048)	1,050,624	activation_41[0]...
conv2d_46 (Conv2D)	(None, 7, 7, 2048)	2,099,200	activation_39[0]...

batch_normalizatio... (BatchNormalizatio...	(None, 7, 7, 2048)	8,192	conv2d_45[0][0]
batch_normalizatio... (BatchNormalizatio...	(None, 7, 7, 2048)	8,192	conv2d_46[0][0]
add_13 (Add)	(None, 7, 7, 2048)	0	batch_normalizat... batch_normalizat...
activation_42 (Activation)	(None, 7, 7, 2048)	0	add_13[0][0]
conv2d_47 (Conv2D)	(None, 7, 7, 512)	1,049,088	activation_42[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv2d_47[0][0]
activation_43 (Activation)	(None, 7, 7, 512)	0	batch_normalizat...
conv2d_48 (Conv2D)	(None, 7, 7, 512)	2,359,808	activation_43[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv2d_48[0][0]
activation_44 (Activation)	(None, 7, 7, 512)	0	batch_normalizat...
conv2d_49 (Conv2D)	(None, 7, 7, 2048)	1,050,624	activation_44[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 7, 7, 2048)	8,192	conv2d_49[0][0]
add_14 (Add)	(None, 7, 7, 2048)	0	batch_normalizat... activation_42[0]...
activation_45 (Activation)	(None, 7, 7, 2048)	0	add_14[0][0]
conv2d_50 (Conv2D)	(None, 7, 7, 512)	1,049,088	activation_45[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv2d_50[0][0]
activation_46 (Activation)	(None, 7, 7, 512)	0	batch_normalizat...
conv2d_51 (Conv2D)	(None, 7, 7, 512)	2,359,808	activation_46[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv2d_51[0][0]
activation_47 (Activation)	(None, 7, 7, 512)	0	batch_normalizat...

conv2d_52 (Conv2D)	(None, 7, 7, 2048)	1,050,624	activation_47[0]...
batch_normalizatio... (BatchNormalizatio...	(None, 7, 7, 2048)	8,192	conv2d_52[0][0]
add_15 (Add)	(None, 7, 7, 2048)	0	batch_normalizat... activation_45[0]...
activation_48 (Activation)	(None, 7, 7, 2048)	0	add_15[0][0]
global_average_poo... (GlobalAveragePool...	(None, 2048)	0	activation_48[0]...
dense (Dense)	(None, 15)	30,735	global_average_p...

Total params: 23,618,447 (90.10 MB)

Trainable params: 23,565,327 (89.89 MB)

Non-trainable params: 53,120 (207.50 KB)

```
In [17]: # Compile the model
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
              loss='categorical_crossentropy',
              metrics=['accuracy'])
```

```
In [18]: # Train the model
history = model.fit(
    train_dataset,
    validation_data=validation_dataset,
    epochs=5 # Adjust epochs as needed
)
```

Epoch 1/5

184/184 ————— 187s 698ms/step - accuracy: 0.1393 - loss: 3.3689 - val
_accuracy: 0.0881 - val_loss: 23.4117

Epoch 2/5

184/184 ————— 103s 557ms/step - accuracy: 0.2058 - loss: 2.5645 - val
_accuracy: 0.2225 - val_loss: 2.4699

Epoch 3/5

184/184 ————— 104s 562ms/step - accuracy: 0.2581 - loss: 2.3371 - val
_accuracy: 0.2771 - val_loss: 2.3595

Epoch 4/5

184/184 ————— 103s 555ms/step - accuracy: 0.2983 - loss: 2.2131 - val
_accuracy: 0.2375 - val_loss: 2.4472

Epoch 5/5

184/184 ————— 105s 569ms/step - accuracy: 0.3221 - loss: 2.1835 - val
_accuracy: 0.1993 - val_loss: 2.7987

Pretrained model and fine tuning

```
In [19]: import tensorflow as tf
from tensorflow.keras.applications import ResNet50
from tensorflow.keras import layers, models

# Set up directory and parameters
train_dir = '/content/train'
img_size = (224, 224)
batch_size = 32
num_classes = 15 # Update this with the actual number of classes in your dataset

# Load the pre-trained ResNet50 model excluding the top layers
base_model = ResNet50(weights='imagenet', include_top=False, input_shape=(224, 224, 3))

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5
94765736/94765736 ————— 5s 0us/step
```

```
In [20]: base_model.summary()
```

Model: "resnet50"

Layer (type)	Output Shape	Param #	Connected to
input_layer_2 (InputLayer)	(None, 224, 224, 3)	0	-
conv1_pad (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer_2[0]...
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalizatio...	(None, 112, 112, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 114, 114, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 56, 56, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 56, 56, 64)	4,160	pool1_pool[0][0]
conv2_block1_1_bn (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2_block1_1_c...
conv2_block1_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_1_b...
conv2_block1_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block1_1_r...
conv2_block1_2_bn (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2_block1_2_c...
conv2_block1_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_2_b...
conv2_block1_0_conv (Conv2D)	(None, 56, 56, 256)	16,640	pool1_pool[0][0]
conv2_block1_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block1_2_r...
conv2_block1_0_bn (BatchNormalizatio...	(None, 56, 56, 256)	1,024	conv2_block1_0_c...
conv2_block1_3_bn (BatchNormalizatio...	(None, 56, 56, 256)	1,024	conv2_block1_3_c...
conv2_block1_add (Add)	(None, 56, 56, 256)	0	conv2_block1_0_b... conv2_block1_3_b...

conv2_block1_out (Activation)	(None, 56, 56, 256)	0	conv2_block1_add_
conv2_block2_1_conv (Conv2D)	(None, 56, 56, 64)	16,448	conv2_block1_out_
conv2_block2_1_bn (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2_block2_1_c_
conv2_block2_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block2_1_b_
conv2_block2_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block2_1_r_
conv2_block2_2_bn (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2_block2_2_c_
conv2_block2_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block2_2_b_
conv2_block2_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block2_2_r_
conv2_block2_3_bn (BatchNormalizatio...	(None, 56, 56, 256)	1,024	conv2_block2_3_c_
conv2_block2_add (Add)	(None, 56, 56, 256)	0	conv2_block1_out_ conv2_block2_3_b_
conv2_block2_out (Activation)	(None, 56, 56, 256)	0	conv2_block2_add_
conv2_block3_1_conv (Conv2D)	(None, 56, 56, 64)	16,448	conv2_block2_out_
conv2_block3_1_bn (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2_block3_1_c_
conv2_block3_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block3_1_b_
conv2_block3_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block3_1_r_
conv2_block3_2_bn (BatchNormalizatio...	(None, 56, 56, 64)	256	conv2_block3_2_c_
conv2_block3_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block3_2_b_
conv2_block3_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block3_2_r_
conv2_block3_3_bn	(None, 56, 56,	1,024	conv2_block3_3_c_

(BatchNormalizatio...	256)		
conv2_block3_add (Add)	(None, 56, 56, 256)	0	conv2_block2_out_ conv2_block3_3_b...
conv2_block3_out (Activation)	(None, 56, 56, 256)	0	conv2_block3_add...
conv3_block1_1_conv (Conv2D)	(None, 28, 28, 128)	32,896	conv2_block3_out...
conv3_block1_1_bn (BatchNormalizatio...	(None, 28, 28, 128)	512	conv3_block1_1_c...
conv3_block1_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block1_1_b...
conv3_block1_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block1_1_r...
conv3_block1_2_bn (BatchNormalizatio...	(None, 28, 28, 128)	512	conv3_block1_2_c...
conv3_block1_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block1_2_b...
conv3_block1_0_conv (Conv2D)	(None, 28, 28, 512)	131,584	conv2_block3_out...
conv3_block1_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block1_2_r...
conv3_block1_0_bn (BatchNormalizatio...	(None, 28, 28, 512)	2,048	conv3_block1_0_c...
conv3_block1_3_bn (BatchNormalizatio...	(None, 28, 28, 512)	2,048	conv3_block1_3_c...
conv3_block1_add (Add)	(None, 28, 28, 512)	0	conv3_block1_0_b... conv3_block1_3_b...
conv3_block1_out (Activation)	(None, 28, 28, 512)	0	conv3_block1_add...
conv3_block2_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block1_out...
conv3_block2_1_bn (BatchNormalizatio...	(None, 28, 28, 128)	512	conv3_block2_1_c...
conv3_block2_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block2_1_b...
conv3_block2_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block2_1_r...

conv3_block2_2_bn (BatchNormalizatio...	(None, 28, 28, 128)	512	conv3_block2_2_c...
conv3_block2_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block2_2_b...
conv3_block2_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block2_2_r...
conv3_block2_3_bn (BatchNormalizatio...	(None, 28, 28, 512)	2,048	conv3_block2_3_c...
conv3_block2_add (Add)	(None, 28, 28, 512)	0	conv3_block1_out...
conv3_block2_out (Activation)	(None, 28, 28, 512)	0	conv3_block2_add...
conv3_block3_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block2_out...
conv3_block3_1_bn (BatchNormalizatio...	(None, 28, 28, 128)	512	conv3_block3_1_c...
conv3_block3_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block3_1_b...
conv3_block3_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block3_1_r...
conv3_block3_2_bn (BatchNormalizatio...	(None, 28, 28, 128)	512	conv3_block3_2_c...
conv3_block3_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block3_2_b...
conv3_block3_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block3_2_r...
conv3_block3_3_bn (BatchNormalizatio...	(None, 28, 28, 512)	2,048	conv3_block3_3_c...
conv3_block3_add (Add)	(None, 28, 28, 512)	0	conv3_block2_out...
conv3_block3_out (Activation)	(None, 28, 28, 512)	0	conv3_block3_add...
conv3_block4_1_conv (Conv2D)	(None, 28, 28, 128)	65,664	conv3_block3_out...
conv3_block4_1_bn (BatchNormalizatio...	(None, 28, 28, 128)	512	conv3_block4_1_c...
conv3_block4_1_relu (Activation)	(None, 28, 28, 128)	0	conv3_block4_1_b...

conv3_block4_2_conv (Conv2D)	(None, 28, 28, 128)	147,584	conv3_block4_1_r...
conv3_block4_2_bn (BatchNormalizatio...	(None, 28, 28, 128)	512	conv3_block4_2_c...
conv3_block4_2_relu (Activation)	(None, 28, 28, 128)	0	conv3_block4_2_b...
conv3_block4_3_conv (Conv2D)	(None, 28, 28, 512)	66,048	conv3_block4_2_r...
conv3_block4_3_bn (BatchNormalizatio...	(None, 28, 28, 512)	2,048	conv3_block4_3_c...
conv3_block4_add (Add)	(None, 28, 28, 512)	0	conv3_block3_out... conv3_block4_3_b...
conv3_block4_out (Activation)	(None, 28, 28, 512)	0	conv3_block4_add...
conv4_block1_1_conv (Conv2D)	(None, 14, 14, 256)	131,328	conv3_block4_out...
conv4_block1_1_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block1_1_c...
conv4_block1_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block1_1_b...
conv4_block1_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block1_1_r...
conv4_block1_2_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block1_2_c...
conv4_block1_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block1_2_b...
conv4_block1_0_conv (Conv2D)	(None, 14, 14, 1024)	525,312	conv3_block4_out...
conv4_block1_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block1_2_r...
conv4_block1_0_bn (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv4_block1_0_c...
conv4_block1_3_bn (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv4_block1_3_c...
conv4_block1_add (Add)	(None, 14, 14, 1024)	0	conv4_block1_0_b... conv4_block1_3_b...
conv4_block1_out	(None, 14, 14,	0	conv4_block1_add...

(Activation)	1024)		
conv4_block2_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block1_out_
conv4_block2_1_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block2_1_c_
conv4_block2_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block2_1_b_
conv4_block2_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block2_1_r_
conv4_block2_2_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block2_2_c_
conv4_block2_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block2_2_b_
conv4_block2_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block2_2_r_
conv4_block2_3_bn (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv4_block2_3_c_
conv4_block2_add (Add)	(None, 14, 14, 1024)	0	conv4_block1_out_ conv4_block2_3_b_
conv4_block2_out (Activation)	(None, 14, 14, 1024)	0	conv4_block2_add_
conv4_block3_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block2_out_
conv4_block3_1_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block3_1_c_
conv4_block3_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block3_1_b_
conv4_block3_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block3_1_r_
conv4_block3_2_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block3_2_c_
conv4_block3_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block3_2_b_
conv4_block3_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block3_2_r_
conv4_block3_3_bn (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv4_block3_3_c_

conv4_block3_add (Add)	(None, 14, 14, 1024)	0	conv4_block2_out_ conv4_block3_3_b...
conv4_block3_out (Activation)	(None, 14, 14, 1024)	0	conv4_block3_add...
conv4_block4_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block3_out...
conv4_block4_1_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block4_1_c...
conv4_block4_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block4_1_b...
conv4_block4_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block4_1_r...
conv4_block4_2_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block4_2_c...
conv4_block4_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block4_2_b...
conv4_block4_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block4_2_r...
conv4_block4_3_bn (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv4_block4_3_c...
conv4_block4_add (Add)	(None, 14, 14, 1024)	0	conv4_block3_out_ conv4_block4_3_b...
conv4_block4_out (Activation)	(None, 14, 14, 1024)	0	conv4_block4_add...
conv4_block5_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block4_out...
conv4_block5_1_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block5_1_c...
conv4_block5_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block5_1_b...
conv4_block5_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block5_1_r...
conv4_block5_2_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block5_2_c...
conv4_block5_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block5_2_b...
conv4_block5_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block5_2_r...

conv4_block5_3_bn (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv4_block5_3_c...
conv4_block5_add (Add)	(None, 14, 14, 1024)	0	conv4_block4_out_ conv4_block5_3_b...
conv4_block5_out (Activation)	(None, 14, 14, 1024)	0	conv4_block5_add...
conv4_block6_1_conv (Conv2D)	(None, 14, 14, 256)	262,400	conv4_block5_out...
conv4_block6_1_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block6_1_c...
conv4_block6_1_relu (Activation)	(None, 14, 14, 256)	0	conv4_block6_1_b...
conv4_block6_2_conv (Conv2D)	(None, 14, 14, 256)	590,080	conv4_block6_1_r...
conv4_block6_2_bn (BatchNormalizatio...	(None, 14, 14, 256)	1,024	conv4_block6_2_c...
conv4_block6_2_relu (Activation)	(None, 14, 14, 256)	0	conv4_block6_2_b...
conv4_block6_3_conv (Conv2D)	(None, 14, 14, 1024)	263,168	conv4_block6_2_r...
conv4_block6_3_bn (BatchNormalizatio...	(None, 14, 14, 1024)	4,096	conv4_block6_3_c...
conv4_block6_add (Add)	(None, 14, 14, 1024)	0	conv4_block5_out_ conv4_block6_3_b...
conv4_block6_out (Activation)	(None, 14, 14, 1024)	0	conv4_block6_add...
conv5_block1_1_conv (Conv2D)	(None, 7, 7, 512)	524,800	conv4_block6_out...
conv5_block1_1_bn (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv5_block1_1_c...
conv5_block1_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block1_1_b...
conv5_block1_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block1_1_r...
conv5_block1_2_bn (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv5_block1_2_c...
conv5_block1_2_relu	(None, 7, 7, 512)	0	conv5_block1_2_b...

(Activation)			
conv5_block1_0_conv (Conv2D)	(None, 7, 7, 2048)	2,099,200	conv4_block6_out_
conv5_block1_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block1_2_r_
conv5_block1_0_bn (BatchNormalizatio...	(None, 7, 7, 2048)	8,192	conv5_block1_0_c_
conv5_block1_3_bn (BatchNormalizatio...	(None, 7, 7, 2048)	8,192	conv5_block1_3_c_
conv5_block1_add (Add)	(None, 7, 7, 2048)	0	conv5_block1_0_b_ conv5_block1_3_b_
conv5_block1_out (Activation)	(None, 7, 7, 2048)	0	conv5_block1_add_
conv5_block2_1_conv (Conv2D)	(None, 7, 7, 512)	1,049,088	conv5_block1_out_
conv5_block2_1_bn (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv5_block2_1_c_
conv5_block2_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block2_1_b_
conv5_block2_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block2_1_r_
conv5_block2_2_bn (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv5_block2_2_c_
conv5_block2_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block2_2_b_
conv5_block2_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block2_2_r_
conv5_block2_3_bn (BatchNormalizatio...	(None, 7, 7, 2048)	8,192	conv5_block2_3_c_
conv5_block2_add (Add)	(None, 7, 7, 2048)	0	conv5_block1_out_ conv5_block2_3_b_
conv5_block2_out (Activation)	(None, 7, 7, 2048)	0	conv5_block2_add_
conv5_block3_1_conv (Conv2D)	(None, 7, 7, 512)	1,049,088	conv5_block2_out_
conv5_block3_1_bn (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv5_block3_1_c_

conv5_block3_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block3_1_b...
conv5_block3_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block3_1_r...
conv5_block3_2_bn (BatchNormalizatio...	(None, 7, 7, 512)	2,048	conv5_block3_2_c...
conv5_block3_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block3_2_b...
conv5_block3_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block3_2_r...
conv5_block3_3_bn (BatchNormalizatio...	(None, 7, 7, 2048)	8,192	conv5_block3_3_c...
conv5_block3_add (Add)	(None, 7, 7, 2048)	0	conv5_block2_out_ conv5_block3_3_b...
conv5_block3_out (Activation)	(None, 7, 7, 2048)	0	conv5_block3_add...

Total params: 23,587,712 (89.98 MB)

Trainable params: 23,534,592 (89.78 MB)

Non-trainable params: 53,120 (207.50 KB)

```
In [21]: # Freeze the base model layers
base_model.trainable = False

# Create a new model on top of the pre-trained base model
inputs = tf.keras.Input(shape=(224, 224, 3))

# Normalization layer
x = layers.Rescaling(1./255)(inputs)

# Optional: Apply augmentation (if augmentation is needed)
# x = Layers.RandomFlip('horizontal')(x)
# x = Layers.RandomRotation(0.1)(x)
# x = Layers.RandomZoom(0.1)(x)

# Pass the inputs through the pre-trained ResNet50 base model
x = base_model(x, training=False)

# Global average pooling to reduce the output to a single vector per image
x = layers.GlobalAveragePooling2D()(x)

# Add a fully connected layer with the number of classes
x = layers.Dense(256, activation='relu')(x)
x = layers.Dropout(0.5)(x) # Dropout to reduce overfitting

# Final output layer with softmax activation for classification
outputs = layers.Dense(num_classes, activation='softmax')(x)
```

```
# Build the model
model = models.Model(inputs, outputs)

model.summary()
```

Model: "functional_2"

Layer (type)	Output Shape	Param #
input_layer_3 (InputLayer)	(None, 224, 224, 3)	0
rescaling (Rescaling)	(None, 224, 224, 3)	0
resnet50 (Functional)	(None, 7, 7, 2048)	23,587,712
global_average_pooling2d_1 (GlobalAveragePooling2D)	(None, 2048)	0
dense_1 (Dense)	(None, 256)	524,544
dropout (Dropout)	(None, 256)	0
dense_2 (Dense)	(None, 15)	3,855

Total params: 24,116,111 (92.00 MB)

Trainable params: 528,399 (2.02 MB)

Non-trainable params: 23,587,712 (89.98 MB)

In [21]:

```
In [22]: # Compile the model
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Load training and validation datasets using image_dataset_from_directory
train_dataset = tf.keras.preprocessing.image_dataset_from_directory(
    train_dir,
    validation_split=0.2,
    subset="training",
    seed=123,
    image_size=img_size,
    batch_size=batch_size
)

validation_dataset = tf.keras.preprocessing.image_dataset_from_directory(
    train_dir,
    validation_split=0.2,
    subset="validation",
    seed=123,
    image_size=img_size,
    batch_size=batch_size
)
```

```

)

# Prefetch data for performance improvement
AUTOTUNE = tf.data.experimental.AUTOTUNE
train_dataset = train_dataset.prefetch(buffer_size=AUTOTUNE)
validation_dataset = validation_dataset.prefetch(buffer_size=AUTOTUNE)

# Train the new layers while keeping the base model frozen
history = model.fit(
    train_dataset,
    validation_data=validation_dataset,
    epochs=5 # Adjust epochs as needed
)

```

Found 7328 files belonging to 15 classes.

Using 5863 files for training.

Found 7328 files belonging to 15 classes.

Using 1465 files for validation.

Epoch 1/5

184/184 ————— 39s 152ms/step - accuracy: 0.0949 - loss: 2.7831 - val_
accuracy: 0.1249 - val_loss: 2.6207

Epoch 2/5

184/184 ————— 29s 111ms/step - accuracy: 0.1180 - loss: 2.6348 - val_
accuracy: 0.1345 - val_loss: 2.5855

Epoch 3/5

184/184 ————— 41s 112ms/step - accuracy: 0.1261 - loss: 2.6056 - val_
accuracy: 0.1686 - val_loss: 2.5652

Epoch 4/5

184/184 ————— 20s 111ms/step - accuracy: 0.1427 - loss: 2.5909 - val_
accuracy: 0.1570 - val_loss: 2.5650

Epoch 5/5

184/184 ————— 41s 110ms/step - accuracy: 0.1454 - loss: 2.5625 - val_
accuracy: 0.1884 - val_loss: 2.5382

Layers fine tuning

```

In [23]: # Load the pre-trained ResNet50 model excluding the top layers
base_model = ResNet50(weights='imagenet', include_top=False, input_shape=(224, 224,

# Unfreeze the last few layers of the base model for fine-tuning
base_model.trainable = True

# Let's set up how many layers we want to fine-tune
# For example, we can choose to fine-tune the last 10 layers of the base model
fine_tune_at = 140 # Unfreeze from this layer onwards

# Freeze all layers before `fine_tune_at`
for layer in base_model.layers[:fine_tune_at]:
    layer.trainable = False

# Create a new model on top of the pre-trained base model with the unfrozen layers
inputs = tf.keras.Input(shape=(224, 224, 3))

# Normalization layer

```

```

x = layers.Rescaling(1./255)(inputs)

# Optional: Apply augmentation (if augmentation is needed)
# x = Layers.RandomFlip('horizontal')(x)
# x = Layers.RandomRotation(0.1)(x)
# x = Layers.RandomZoom(0.1)(x)

# Pass the inputs through the pre-trained ResNet50 base model
x = base_model(x, training=False) # Keep training=False here

# Global average pooling to reduce the output to a single vector per image
x = layers.GlobalAveragePooling2D()(x)

# Add a fully connected layer with the number of classes
x = layers.Dense(256, activation='relu')(x)
x = layers.Dropout(0.5)(x) # Dropout to reduce overfitting

# Final output layer with softmax activation for classification
outputs = layers.Dense(num_classes, activation='softmax')(x)

# Build the model
model = models.Model(inputs, outputs)

```

```
In [24]: model.summary()
```

Model: "functional_3"

Layer (type)	Output Shape	Param #
input_layer_5 (InputLayer)	(None, 224, 224, 3)	0
rescaling_1 (Rescaling)	(None, 224, 224, 3)	0
resnet50 (Functional)	(None, 7, 7, 2048)	23,587,712
global_average_pooling2d_2 (GlobalAveragePooling2D)	(None, 2048)	0
dense_3 (Dense)	(None, 256)	524,544
dropout_1 (Dropout)	(None, 256)	0
dense_4 (Dense)	(None, 15)	3,855

Total params: 24,116,111 (92.00 MB)

Trainable params: 15,506,447 (59.15 MB)

Non-trainable params: 8,609,664 (32.84 MB)

```

In [25]: # Compile the model again after unfreezing layers (lower learning rate for fine-tune)
model.compile(optimizer=tf.keras.optimizers.Adam(learning_rate=1e-5), # Lower learning rate
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Fine-tune the model

```

```
fine_tune_history = model.fit(  
    train_dataset,  
    validation_data=validation_dataset,  
    epochs=5 # Increase epochs for fine-tuning  
)
```

Epoch 1/5

184/184 ————— 59s 214ms/step - accuracy: 0.1012 - loss: 2.7631 - val_
accuracy: 0.1017 - val_loss: 2.6804

Epoch 2/5

184/184 ————— 27s 145ms/step - accuracy: 0.2164 - loss: 2.4402 - val_
accuracy: 0.2546 - val_loss: 2.3753

Epoch 3/5

184/184 ————— 41s 148ms/step - accuracy: 0.2907 - loss: 2.2690 - val_
accuracy: 0.3276 - val_loss: 2.2236

Epoch 4/5

184/184 ————— 41s 148ms/step - accuracy: 0.3307 - loss: 2.1158 - val_
accuracy: 0.3420 - val_loss: 2.1203

Epoch 5/5

184/184 ————— 41s 146ms/step - accuracy: 0.3897 - loss: 1.9701 - val_
accuracy: 0.3706 - val_loss: 2.0555

```
In [1]: a, b, *c, (d, e) = [1,2,(5, 6)]  
        print(a, b, c, d, e)
```

1 2 [] 5 6

```
In [ ]:
```